

Telco Customer Churn

Business Insights & KPI Framework for Customer Retention

1. Executive Summary

The dataset covers 7,043 customers with an overall churn rate of 26.5% (1,869 churned customers) against a retention rate of 73.5%. Churned customers pay more per month on average (\$74.44) than retained customers (\$61.27), yet leave earlier — average tenure at churn is 17.98 months versus 37.57 months for retained customers. This combination — higher-paying customers leaving faster — means churn is disproportionately expensive: the business is losing an estimated \$139,128 in monthly recurring revenue (~\$1.67M annualized) to churn, against a retained base of \$317,011/month (~\$3.80M annualized). The single largest structural driver of churn is contract type: month-to-month customers churn at 42.7%, more than 15x the two-year contract rate (2.8%), and account for 88.6% of all churn.

2. Key Business Insights from EDA

2.1 Revenue & Tenure

- Churned customers have ~22% higher MonthlyCharges than retained customers (\$74.44 vs \$61.27) — the business is losing its more valuable, higher-ARPU customers at a disproportionate rate.
- Churned customers have less than half the tenure of retained customers (17.98 vs 37.57 months) and roughly 40% lower lifetime TotalCharges (\$1,531.80 vs \$2,555.34) — most value is lost before a customer relationship matures.
- Churn risk is heavily front-loaded: a large share of cancellations happen in month 1, with elevated churn persisting through months 2–5, then tapering steadily as tenure increases — classic early-lifecycle attrition.
- MonthlyCharges above ~\$70 mark a clear inflection point where churn probability rises sharply, suggesting price sensitivity or a value-for-money gap at higher price tiers.

2.2 Contract, Billing & Payment

- **Month-to-month contracts churn at 42.7% vs 11.3% (one-year) and 2.8% (two-year) — contract length is the strongest lever the business has over churn.**
- Electronic check payers show a materially higher churn share than automatic payment methods: of 2,365 customers paying by electronic check, 1,071 churned — a 45.3% churn rate within that segment, and 57.3% of all churn company-wide.
- PaperlessBilling customers churn more than paper-billing customers, suggesting either weaker engagement with digital billing communications or that paperless enrollment correlates with less-committed customer segments.

2.3 Services & Support

- Customers without OnlineSecurity, OnlineBackup, DeviceProtection, or TechSupport churn substantially more (~71.6% average) than those with these services — these are retention-anchoring add-ons, not just upsell revenue.
- Fiber optic internet customers churn more than DSL customers, despite Fiber being the premium/higher-revenue product — points to a service quality, reliability, or price-perception gap specific to Fiber.
- StreamingTV/StreamingMovies show no protective effect on their own — churn happens regardless of streaming subscription, indicating streaming is not a strong retention driver by itself.

2.4 Customer Profile

- Customers without a partner or dependents churn more — the model treats "living independently" as a segment worth targeted retention outreach (nearly 3 in 4 churned customers fit this profile).
- SeniorCitizens are a smaller share of the base (1,142 of 7,043) but churn at a notably higher rate within their segment: 476 of 1,142 churned (41.7%) — despite being fewer in number, they represent a high-risk, high-service-expectation segment.

3. KPI Dashboard: Definitions & Current Baseline

Formulas and current baseline values for the six KPIs selected, computed from the full 7,043-customer dataset (MonthlyCharges used as the revenue proxy for recurring monthly revenue).

KPI	Formula	Current Value	What it tells you
Average Revenue Per Customer (ARPU)	Total Monthly Charges ÷ Total Customers	\$64.76 / month	Overall pricing/monetization level
Churned Customers by Contract	Count of churned customers grouped by Contract type	MTM: 1,655 (88.6%) · 1-yr: 166 (8.9%) · 2-yr: 48 (2.6%)	Where churn risk concentrates
Churned Revenue (monthly, lost)	Σ MonthlyCharges of churned customers	\$139,128 / month (~\$1.67M / year)	Revenue actively being lost to churn
Retained Revenue (monthly)	Σ MonthlyCharges of retained customers	\$317,011 / month (~\$3.80M / year)	Revenue base currently secured
Retention Rate	Retained Customers ÷ Total Customers	73.5%	Share of base successfully kept
Churn Rate	Churned Customers ÷ Total Customers	26.5%	Share of base lost in the period

3.1 Churn by Contract — Detail

Contract Type	Total Customers	Churned	Churn Rate	Share of All Churn
Month-to-month	3,875	1,655	42.7%	88.6%
One year	1,473	166	11.3%	8.9%
Two year	1,695	48	2.8%	2.6%

4. KPI-Linked Retention Recommendations

4.1 Reduce Churn Rate / Raise Retention Rate

- Migrate month-to-month customers to annual contracts through a structured incentive ladder (e.g., a modest discount or a free add-on service for a 1-year commitment) — this is the single highest-leverage action given the 42.7% vs 2.8% churn gap.

- Build an early-lifecycle "first 90 days" retention program (proactive check-ins, onboarding support, first-bill review) targeting the front-loaded churn spike in months 1–5.
- Create a churn-risk score (using tenure, contract type, payment method, and add-on service count) to trigger proactive outreach before cancellation, rather than reacting after the fact.

4.2 Protect Churned Revenue / Grow Retained Revenue

- Prioritize save-desk and win-back efforts on high-MonthlyCharges customers on month-to-month contracts — this segment drives the largest revenue-at-risk per customer.
- Bundle OnlineSecurity, OnlineBackup, DeviceProtection, and TechSupport into entry-level packages rather than treating them as pure upsells — the data shows these services correlate strongly with retention, not just incremental revenue.
- Audit Fiber optic service quality/pricing perception specifically — losing premium-ARPU customers on the premium product is a direct hit to both churned revenue and long-term ARPU growth.

4.3 Lift Average Revenue Per Customer (ARPU) Sustainably

- Since churned customers already have higher ARPU than retained customers, raising prices without fixing retention drivers will likely worsen churn — pair any ARPU-growth initiative with the retention actions above.
- Grow ARPU through service attach (security/support bundles, which are retention-positive) rather than base-price increases, especially above the ~\$70/month threshold where churn risk rises sharply.

4.4 Fix Payment & Billing Friction

- Incentivize migration from Electronic check to automatic payment methods (bank transfer/credit card autopay) — e.g., a small statement credit — since manual payment methods show materially higher churn association.
- Review paperless billing communications (clarity, timing, spam-folder risk) to ensure the digital billing experience isn't itself a churn driver.

5. Suggested Ongoing Tracking Cadence

- Track Churn Rate and Retention Rate monthly at the overall level, and by Contract type, Payment Method, and Internet Service segment — the segments where this analysis found the largest gaps.
- Track Churned Revenue and Retained Revenue monthly in dollar terms (not just customer counts) so the finance and retention teams share one number for "revenue at risk."
- Track ARPU split by churned vs. retained cohorts each month — a widening gap (churned ARPU pulling further above retained ARPU) is an early warning that the business is losing its best customers.